No. of Functional Features Included in the Solution

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Login & Authentication	Secure role-based access for Admins, Data Analysts, and Viewers.	Security / Auth	Done	0.5
2	Spreadsheet Drag-and-Drop Ingestion	Ingests Excel (.xlsx) and CSV files into temporary staging tables.	User Interface & Ingestion	Done	0.5
3	Pre-Import File Validation	Checks file format, maximum file size, and missing headers before processing.	Backend Validation	Done	0.5
4	Dynamic Field Mapping Engine	Auto-suggests and manually maps source spreadsheet columns to target schema fields.	Transform Map Core	Done	1.0
5	Coalesce Duplicate Prevention Rule	Enforces matching key rules to update existing records instead of creating duplicates.	Core Transformation Logic	Done	1.0
6	Batch Processing Execution	Transforms and loads raw staging data into target database tables in batches.	Database Writer	Done	0.5

7	Real-Time Log & Error Dashboard	Displays import status, row counts, skipped records, and row-level error reports.	UI & Reporting	Done	0.5
8	Failed Row Interactive Retry	Allows users to edit invalid row data in a grid UI and re-run transformation.	UI & Logic	In progress	0.5

Feature Summary:

Metric	**Count / Value**
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	**Category**	**Features in Category**	**Example Features**
1	User Interface (UI)	3	Drag-and-drop file uploader, dynamic field mapping canvas, real-time log dashboard.
2	Backend / Logic	2	Field auto-mapping algorithm, Coalesce rule duplicate-matching engine.
3	Database / Storage	1	Import Set staging tables and target database record synchronization.
4	API / Integration	1	Spreadsheet parsing service integration (pandas / openpyxl).
5	Security / Authentication	1	JWT-based authentication and role-based access control (RBAC).